Experiment -5

Python Program To Missionaries and Cannibals Problem

Code:

```
from collections import deque


def valid(m, c):
    return (0 <= m <= 3 and 0 <= c <= 3 and
            (m == 0 or m >= c) and
            (m == 3 or 3-m >= 3-c))


def solve():
    start = (3, 3, 0)
    goal = (0, 0, 1)

    queue = deque([(start, [start])])
    visited = set()

    moves = [(1, 0), (2, 0), (0, 1),
             (0, 2), (1, 1)]

    while queue:
        state, path = queue.popleft()

        if state == goal:
            return path

        if state in visited:
```

```
            continue

        visited.add(state)
        m, c, boat = state

        for dm, dc in moves:
            if boat == 0:
                new = (m-dm, c-dc, 1)
            else:
                new = (m+dm, c+dc, 0)

            if valid(new[0], new[1]) and new not in visited:
                queue.append((new, path + [new]))

    return None

solution = solve()

if solution:
    for state in solution:
        print(state)
else:
    print("No solution")
```

```python
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            (m == 0 or m >= c) and
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    return None

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if solution:
    for state in solution:
        print(state)
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```

```
(3, 3, 0)
(3, 1, 1)
(3, 2, 0)
(3, 0, 1)
(3, 1, 0)
(1, 1, 1)
(2, 2, 0)
(0, 2, 1)
(0, 3, 0)
(0, 1, 1)
(1, 1, 0)
(0, 0, 1)


...Program finished with exit code 0
Press ENTER to exit console.
```